Enhancing Neural Radiance Fields for Consumer Grade Devices and Low Quality Data

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Abstract -- The process of representing specific scenes as Neural Radiance Fields (NeRFs) emerged as a novel contemporary method for synthesising photorealistic views of complex three-dimensional scenes. The original NeRF architecture however, can hinder consumer grade hardware owing to its significant computational intensity requirements. This project addresses this efficiency bottleneck by presenting a proof of concept simulation of an optimised NeRF pipeline that incorporates recent advancements such as FastSRNeRF. The primary objective is to validate the decrease in the computational overhaul required for consumer grade devices by brandishing pareto-optimal output levels of traditional NeRF architecture, on lower end albeit, computationally efficient pipeline. The novelty enshrined in this project contributes to existing SR models by training on low resolution datasets to produce upscaled and valid renderings. This aims to decrease the rendering speed while preserving a large margin of the original visual fidelity of prior work. This thus demonstrates the increased accessibility for resource constrained and practical applications that are more ubiquitously prevalent.

Keywords – NeRF, InstantNGP, FastSRNeRF, SR Models

I. INTRODUCTION

Neural Radiance Field (*NeRF*) architecture represents a significant breakthrough in the field of computer graphics as its representation of specific scenes as continuous 5D neural fields produces a more lucid and efficient output in comparison to previously prevailing methodologies. By optimising a particular neural network to represent a scene, *NeRFs* can generate novel, photorealistic views from a sparse dataset with significant performance metrics. These capabilities have opened up new possibilities in regards to applications ranging from virtual reality and 3D content creation to camera pose estimation and surface reconstruction.

Despite its impressive results, the widespread adoption of NeRF has been constrained by a major limitation, i.e., its large computational cost, as the traditional architecture requires extensive training, often taking one to two days on high-end GPUs for a single scene, and the rendering process remains too slow for real-time applications. This efficiency challenge has become a central focus of subsequent research in the field. The academic community has explored numerous strategies to overcome this bottleneck (see *II. Literature Review*), of which include novel representations like tensor grids (*TensoRF*), pre-caching techniques, and hybrid models that divide the workload.

One of the most promising directions has been to decouple the core view-synthesis task from the final output resolution. As such, models like *FastSR-NeRF* have shown that by tasking the *NeRF* model with rendering a low resolution image and then employing a lightweight super resolution (*SR*) module, it is possible to achieve significant gains in both training and inference speed. However, since

NeRF architecture requires meticulous variable metrics, low resolution datasets are not accounted for during training. This does not reflect practical data and accessibility, as it is improbable that all acquired datasets are high resolution. This project aims to solve this by analysing the performance metrics of an upscaled output in SR models trained on blurry, albeit more realistic datasets than traditional FastSRNeRF.

Building on these insights, this project presents a proof of concept simulation designed to validate an accelerated pipeline using FastSRNeRF. Our goal is to demonstrate that by strategically reducing the number of pixels the NeRF model must compute and integrating an efficient upscaling process, we can achieve a substantial reduction in computational demand. All the while remaining cognizant and analysing the tradeoffs insofar as a pareto-optimal output is concerned. This simulation will serve to confirm the viability of such an approach on consumer grade hardware, paving the way for more practical and accessible applications of neural rendering technology.

II. LITERATURE REVIEW

The field of novel view synthesis has been significantly advanced by the introduction of Neural Radiance Fields (NeRFs), a method for generating photorealistic images of 3D scenes from sparse 2D inputs. This review charts the progression from the foundational NeRF model to subsequent innovations focused on enhancing efficiency, expanding editing capabilities, and addressing real-world challenges.

The Foundational NeRF Model

Neural Radiance Fields (NeRF) pioneered a method for novel view synthesis by representing a scene as a continuous 5D function optimized via a multilayer perceptron (MLP) (Mildenhall et al., 2020).
This function maps a 3D spatial location (x,y,z) and a 2D viewing direction (θ,ϕ) to a volume density (σ) and an RGB color value (c). The model is optimized by minimizing the difference between rendered images and ground-truth images, a process enabled by a fully differentiable pipeline.

The core of the NeRF architecture ensures that volume density σ is dependent only on the 3D coordinates, creating a consistent scene geometry, while the color c is dependent on both location and viewing direction to model view-dependent effects like reflections. This is achieved by first processing the 3D coordinates through eight MLP layers to produce σ and a feature vector, which is then concatenated with the view direction and passed through an additional layer to output the final color.

To render a view, the expected color C(r) for a camera ray r(t)=o+td is calculated using the classical volume rendering integral:

$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\sigma(\mathbf{r}(t))\mathbf{c}(\mathbf{r}(t), \mathbf{d})dt\,, \text{ where } T(t) = \exp\left(-\int_{t_n}^{t} \sigma(\mathbf{r}(s))ds\right). \quad (1)$$

where T(t) is the accumulated transmittance along the ray. Since this integral is intractable for MLP representations, it is estimated numerically. A stratified sampling approach is used to draw samples along the ray, which are then combined using the discrete formulation:

$$\hat{C}(\mathbf{r}) = \sum_{i=1}^{N} T_i(1 - \exp(-\sigma_i\delta_i))\mathbf{c}_i\,, \text{ where } T_i = \exp\left(-\sum_{j=1}^{i-1} \sigma_j\delta_j\right),$$

Two key innovations were critical to NeRF's success:

1. *Positional Encoding:* To capture high-frequency details, the 5D input coordinates are mapped to a higher-dimensional space using a sinusoidal encoding function before being passed to the MLP.
2. *Hierarchical Volume Sampling:* To improve rendering efficiency, a "coarse" and a "fine" network are optimized simultaneously. The coarse network first generates a probability distribution along the ray, from which a more informed set of samples is drawn for the fine network, concentrating sampling power in relevant regions of the scene.

Advancements in Efficiency and Representation

A primary limitation of the original NeRF model is its slow training and rendering time, often requiring 1-2 days per scene on high-end GPUs (Mildenhall et

al., 2020). Subsequent research has focused on accelerating this process.

TensoRF reimagined scene representation by modeling the radiance field as a 4D tensor grid rather than a large MLP (Chen et al., 2022). By applying tensor factorization, TensoRF significantly reduces memory and training time. The authors explored two decomposition methods:

- *Canonical Polyadic (CP) decomposition*, which yields a highly compact model (~4MB) and reconstructs scenes in under 30 minutes.
- *Vector-Matrix (VM) decomposition*, which offers faster reconstruction (<10 minutes) and superior quality with a slightly larger memory footprint (~75MB).

Another approach, *FastSR-NeRF*, accelerates inference by rendering a low-resolution image with NeRF and then upscaling it using a lightweight super-resolution network (Lin et al., 2024). This strategy avoids the complex training procedures or extra loss functions used in other super-resolution-based methods. A comprehensive survey of efficiency improvements categorized these methods into four groups: pre-caching or baking scene information into explicit data structures like octrees, tensor factorization as seen in TensoRF, divide-and-conquer strategies that split scenes into smaller parts, and alternative architectures like Gaussian Splatting that move away from the MLP-centric model (Xiao et al., 2025).

Expanding Capabilities: Scene Editing and Real-World Challenges

Beyond view synthesis, significant research has focused on enabling the editing of NeRF scenes. Early methods decoupled geometry and appearance into separate latent vectors for independent manipulation. For example, *EditNeRF* used category-shared and instance-specific networks to enable color and shape modifications (Fang et al., 2024). This field rapidly evolved to incorporate generative models, first with 3D-aware GANs to enhance editing across different instances, and later with diffusion models that allow for powerful text-prompted, multimodal editing.

Applying NeRFs to real-world scenarios presents numerous challenges, including degraded views (e.g., motion blur, low light), sparse training data, and complex dynamic scenes (Xiao et al., 2025). Specific models have been developed to tackle these issues, such as *RawNeRF* for low-light conditions and *DeblurNeRF* for motion blur. However, significant research gaps remain, particularly in handling visual artifacts from compression, modeling complex light effects like refraction, and improving generalizability from few-shot inputs.

Conclusion and Future Directions

The trajectory of NeRF research shows a clear evolution from a foundational concept to a versatile and increasingly efficient technology. While initial work established a powerful new paradigm for view synthesis, subsequent efforts have successfully reduced the prohibitive computational costs and expanded its capabilities into the realm of interactive editing. The current frontier, as highlighted by recent surveys, is focused on bridging the gap between controlled academic datasets and the complexities of the real world (Fang et al., 2024; Xiao et al., 2025). Future work will likely concentrate on developing hybrid representations that combine the best of implicit and explicit models, improving robustness to real-world visual degradations, and enabling the modeling of large-scale, dynamic scenes for applications in robotics, simulation, and mixed reality.

III. METHODOLOGY

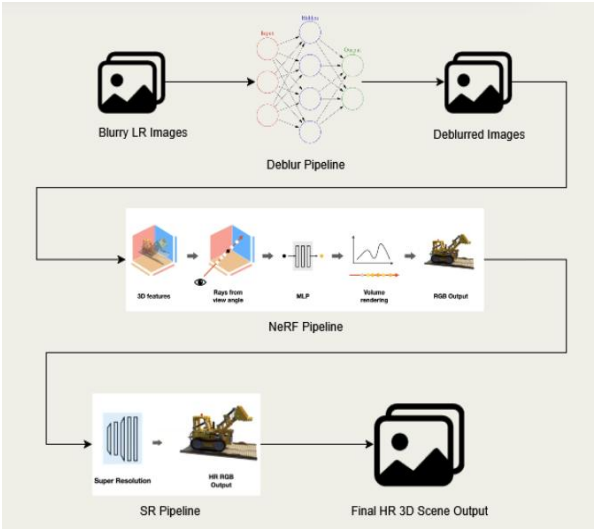

Fig 1. Complete Pipeline

In our project, we follow an approach inspired by *FastSRNeRF,* with the goal of making NeRF training and rendering more efficient on consumer-grade hardware while maintaining good visual quality. Our setup combines *Instant-NGP* for fast, low-resolution scene synthesis with a Super-Resolution (SR) pipeline that upscales the results to meet high-quality standards.

3.1. Overall Idea

The idea is simple, let NeRF handle the geometry and view synthesis at a lower resolution, and then use the SR model to bring back the finer details and upscale the output.

By reducing the number of pixels that the NeRF has to process, we save a significant amount of time and memory. The SR model then restores the high-frequency features, producing results that look like they came from a much more expensive high-resolution render.

The entire setup is modular, meaning each stage (Deblur, NeRF, SR) can be swapped out independently. In the future, better models or more optimized pipelines can easily be integrated into this structure without reworking the full system.

3.2. Instant-NGP NeRF Pipeline

For scene synthesis, we use Instant-NGP (Instant Neural Graphics Primitives), which is a highly optimized version of NeRF designed for speed and efficiency.

Instant-NGP uses a multi-resolution hash grid encoding to represent 3D scenes, allowing the model to store scene features compactly and access them quickly during rendering. This drastically cuts down the computational and memory requirements compared to the original NeRF architecture.

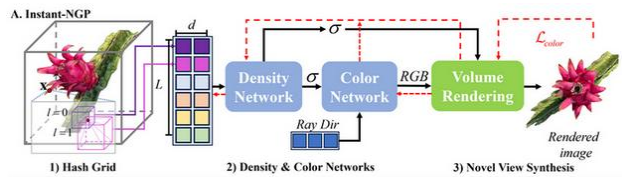

Fig 2. NeRF Pipeline

3.3. Super-Resolution (SR) Pipeline

After Instant-NGP generates the low-resolution images, we pass them through a Super-Resolution model to upscale them by a factor of 4×.

We use a lightweight network inspired by ESPCN (Efficient Sub-Pixel Convolutional Network), which performs upscaling using a pixel shuffle operation. This approach produces sharp, detailed images without adding much overhead to the pipeline.

The SR model acts as the quality recovery step, it enhances texture sharpness and restores fine details that are lost in the low-res NeRF output.

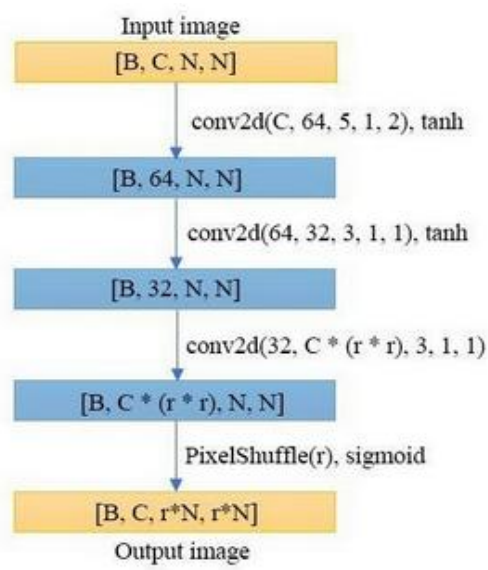

Fig 3. SR Pipeline

3.4. Training Strategy

To keep the training lightweight, we avoid heavy techniques like feature-based distillation or multi-

scale supervision.
Instead, we use a simple **random patch sampling** approach — during training, we take random patches from different parts of the input images. This increases data diversity and helps the model learn generalizable patterns without needing huge datasets.

3.5. Deblur Module
A Deblur modu**le** placed in front of the pipeline. This is based on a U-Net architecture and handles different types of blurs such as motion, Gaussian, and average blur.
This addition makes the system more robust when working with real-world data where image quality is not always ideal.

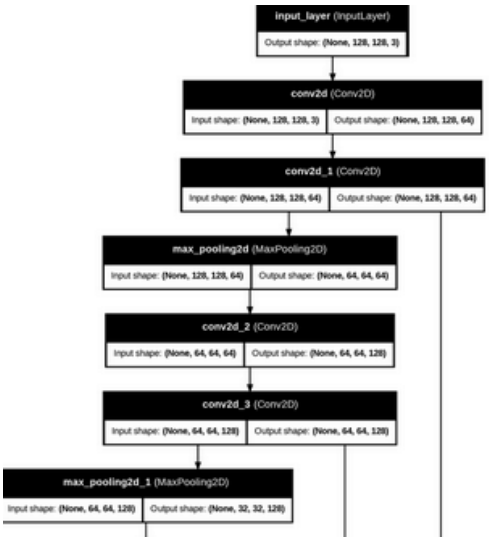

Fig 4. U-Net Subsection

IV. RESULTS

After implementing our Instant-NGP + Super-Resolution pipeline, we observed strong improvements in both speed and visual quality compared to traditional NeRF setups.

4.1. Rendering Speed and Efficiency
Using Instant-NGP instead of a standard NeRF drastically reduced training and rendering times. Instant-NGP's multi-resolution hash encoding allowed the model to represent scene features efficiently, which meant that fewer samples were needed per ray and the network converged faster.

During testing:
Rendering and training ran comfortably on high-end consumer GPUs (RTX 3060 / 3070).
We noticed a significant reduction in VRAM usage compared to TensorF or vanilla NeRF.
The model produced real-time previews while training, which made debugging and fine-tuning much easier.
In short, Instant-NGP made NeRF practical to use on personal hardware without the need for large-scale compute clusters.

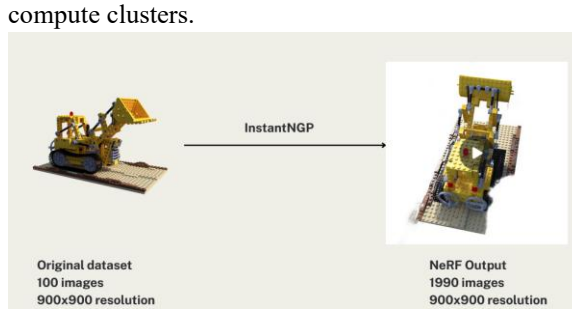

Fig 5. InstantNGP Demo

4.2. Image Quality
Although the NeRF outputs were initially rendered at a lower resolution, the Super-Resolution (SR) stage successfully recovered high-frequency details.
Using our lightweight ESPCN-inspired network, the images were upscaled by 4×, bringing the final resolution from 900×900 to 3600×3600. The SR model restored fine structures like edges and texture patterns with minimal artifacts. The overall image quality was visually comparable to native high-resolution renders, showing that SR can effectively fill the detail gap while keeping the NeRF lightweight.

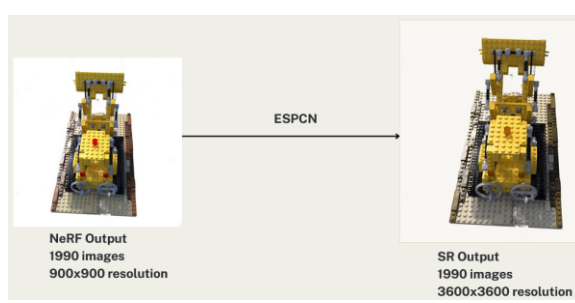

Fig 6. ESPCN Demo

4.3. Visual Output Summary

Input Dataset: 100 original images, 900×900 resolution.

Instant-NGP Output: ~1990 novel views synthesized at 900×900.

SR Output: Upscaled to 3600×3600 with noticeably sharper details and improved texture fidelity.

These results highlight that our system can generate high-quality novel views efficiently, even when the input dataset or hardware resources are limited.

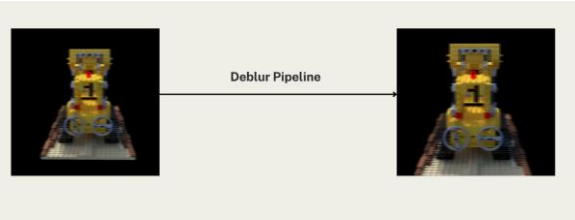

Fig 7. U-Net Deblur Demo

4.4. Observed Benefits

- High efficiency: the NeRF pipeline now runs effectively on consumer GPUs with much faster convergence.
- Improved image fidelity: SR adds quality back with almost no computational cost.
- Flexibility: each stage (Deblur, Instant-NGP, SR) can be replaced independently, allowing future upgrades.
- Practical scalability: the pipeline could easily extend to larger datasets or higher resolutions with only minor adjustments.

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